Project: Worked examples
Title: Mechanism kinematics: four-bar, slider-crank, cam
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1. Scope and method

A single-degree-of-freedom planar mechanism driven at constant speed: a four-bar linkage or an offset slider-crank (both circuits, position, velocity and acceleration over a sweep of the input angle, Grashof class, transmission angle against a minimum, toggle and dead-centre positions, loop-closure residual, input torque by virtual work), or a cam with a translating follower. Rigid links, no clearance, no friction. Four-bar: $a\cdot e^{j\theta_2} + b\cdot e^{j\theta_3} = d + c\cdot e^{j\theta_4}$; $\ell = |A - O_4|$, $\cos\gamma = (c^2 + \ell^2 - b^2)/(2c\ell)$, $\theta_4 = \varphi \pm \gamma$ (φ the direction $O_4 \to A$; open: −, crossed: +); θ_3 from B − A Four-bar velocity and acceleration: differentiated loop, 2×2 system with $\det = b\cdot c\cdot\sin(\theta_3 - \theta_4)$; $\omega_3 = \omega_2\cdot a\cdot\sin(\theta_4 - \theta_2)/(b\cdot\sin(\theta_3 - \theta_4))$, $\omega_4 = \omega_2\cdot a\cdot\sin(\theta_3 - \theta_2)/(c\cdot\sin(\theta_3 - \theta_4))$; α from the second derivative at constant ω_2 Grashof: $s + l < p + q$ (crank-rocker, double-crank or double-rocker by the shortest link), = at the change point, > non-Grashof; transmission angle $\cos\mu = (b^2 + c^2 - \ell^2)/(2bc)$ folded to 0 ... 90°; toggle where $\ell = b + c$ or $|b - c|$ Slider-crank: $x = r\cdot\cos\theta + \sigma\cdot\sqrt{l^2 - (r\cdot\sin\theta - e)^2}$ (σ = +1 open, −1 crossed); $dx/d\theta = -r\cdot\sin\theta - r\cdot\cos\theta\cdot(r\cdot\sin\theta - e)/(x - r\cdot\cos\theta)$; dead centres where crank and rod are collinear; $\mu = 90° - |$rod angle to the slide line$|$ Input torque by virtual work: $T_{in}\cdot\omega_{in} = \Sigma F\cdot v$; four-bar $T_{in} = (T_L + J\cdot\alpha_4)\cdot\omega_4/\omega_2$; slider $T_{in} = (m\cdot a + F_L)\cdot dx/d\theta$; RMS torque and mean power over the sweep by Simpson's rule Forces: four-bar tangential pin force $P = (T_L + J\cdot\alpha_4)/c$, coupler (and crank-pin) force $P/\sin\mu$; slider slide-direction force $F_x = m\cdot a + F_L$, rod force $F_x/\cos\theta_3$, guide reaction $F_x\cdot\tan\theta_3$; cam contact normal $F/\cos\varphi$ Assembly limits: four-bar toggles at $\cos\theta_2 = (d^2 + a^2 - \ell^2)/(2ad)$, $\ell = b + c$ or $|b - c|$; slider $\sin\theta = (e \pm l)/r$. The sweep is cut at them and, driven at constant speed into one, the unbounded peaks are not reported Loop closure: $|a\cdot e^{j\theta_2} + b\cdot e^{j\theta_3} - c\cdot e^{j\theta_4} - d|$ (slider: $|r\cdot e^{j\theta} + l\cdot e^{j\theta_3} - (x + j\cdot e)|$) asserted below 1e-9 mm at every step Cam, per segment with $x = (\theta - \theta_0)/\beta$: rise $s = s_0 + h\cdot f(x)$, return $s = s_0 - h\cdot f(x)$; cycloidal $f = x - \sin(2\pi x)/(2\pi)$; 3-4-5 $f = 10x^3 - 15x^4 + 6x^5$; 4-5-6-7 $f = 35x^4 - 84x^5 + 70x^6 - 20x^7$; modified trapezoid from its acceleration pieces ($C_v = 2$, $C_a = 8\pi/(\pi + 2)$, $C_j = 32\pi^2/(\pi + 2)$) Cam SVAJ at constant ω: $v = \omega\cdot s'$, $a = \omega^2\cdot s''$, $j = \omega^3\cdot s'''$ ($' = d/d\theta$); peak coefficients $C_v = \max f'$, $C_a = \max|f''|$, $C_j = \max|f'''|$, so $v_{pk} = C_v\cdot h\cdot\omega/\beta$ Pressure angle (roller, offset e, $R_p = R_b + R_r$): $\varphi = \arctan[(s' - e)/(\sqrt{R_p^2 - e^2} + s)]$, which for $e = 0$ is $\arctan[s'/(R_p + s)]$; the base circle that holds $\varphi \le \varphi_{max}$ has $R_{p,req} = \sqrt{s_{o,req}^2 + e^2}$ with $s_{o,req} = \max(|s' - e|/\tan\varphi_{max} - s)$ Radius of curvature of the pitch curve $P(\theta) = \mathrm{Rot}(-\theta)\cdot(e, \sqrt{R_p^2 - e^2} + s)$: $\rho = |P'|^3/(P' \times P'')$, convex where positive; undercut where $\rho_{min} \le R_r$ (FAIL), none where the curve is nowhere convex; flat-faced follower: signed profile radius $\rho = R_b + s + s''$, the minimum of which must stay above zero (FAIL at or below) and the face must reach $|s'|$; the cam turns counter-clockwise with the follower axis at +x offset e Cam follower force $F = m\cdot a + k\cdot s + F_{pre} + F_{ext}$ (the follower leaves the cam where $F < 0$); input torque by virtual work $T = F\cdot s'$ ($T\cdot\omega = F\cdot v$)

Ref: Norton, Design of Machinery Ch. 4, 6, 7 — Position, velocity and acceleration analysis; transmission angle; Grashof condition
Ref: Uicker, Pennock and Shigley, Theory of Machines and Mechanisms Ch. 4–6 — Loop-closure position analysis; velocity and acceleration of planar linkages

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Kind of mechanism	Four-bar		spec: Carton line PKG-LINE-7 §3.2 tucker station rev C	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
circuit	Assembly circuit (four-bar, slider-crank): open is $\theta_4 = \varphi - \gamma$ (slider on the +x side of the crank pin), crossed the mirror branch	Open		drawing: PKG-TUCK-02 rev B	any
L1	Ground link d, O_2 to O_4 (four-bar)	150	mm	drawing: PKG-TUCK-02 rev B	1 … 10000 mm
L2	Crank length a (four-bar) or r (slider-crank)	45	mm	drawing: PKG-TUCK-02 rev B	1 … 10000 mm
L3	Coupler length b (four-bar) or connecting-rod length l (slider-crank)	140	mm	drawing: PKG-TUCK-02 rev B	1 … 10000 mm
L4	Rocker length c (four-bar)	100	mm	drawing: PKG-TUCK-02 rev B	1 … 10000 mm
theta_start	Input angle at the first row of the table	0	deg	spec: Carton line PKG-LINE-7 §3.2: crank zero at the home position rev C	-720 … 720 deg
theta_range	Swept range of the input angle (360° for a full turn)	360	deg	spec: Carton line PKG-LINE-7 §3.2: one cycle rev C	1 … 720 deg
steps	Intervals N of the table (N + 1 rows, a whole number, at most 3600; the peaks do not depend on it)	72		assumption: 5° table step.	4 … 720
n_in	Input speed (constant)	60	rpm	spec: Carton line PKG-LINE-7 §3.2: 60 cartons/min rev C	0.1 … 30000 rpm
mu_req	Minimum transmission angle required (linkages)	40	deg	assumption: Default minimum transmission angle; set it from your design rule.	0 … 90 deg
load_dir	External load: constant direction, or always opposing the output motion	Opposes motion		assumption: The flap resists the blade in whichever direction it moves.	any
T_load	External load torque on the output link (four-bar). Positive acts in the $-\theta_4$ direction, resisting positive ω_4; with 'Opposes motion' its sign is applied against the actual direction of ω_4 (a negative value then assists)	4	N*m	assumption: Example flap-folding torque at the rocker shaft; measure it on the station.	-10000 … 10000 N*m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
J_out	Moment of inertia of the output link about its pivot (four-bar)	0.012	kg*m^2	assumption: Example: rocker arm and blade about the rocker pivot; take it from the CAD mass properties.	0 ... 1000 kg*m^2

3. Assumptions

A1. Rigid links, a planar mechanism with one degree of freedom, no joint clearance and no friction; the input turns at a constant speed. (Addendum B §B6 M9 limits.)
A2. The position is the closed-form solution of the vector loop; the open and crossed circuits are the two branches ($\theta_4 = \varphi - \gamma$ and $\varphi + \gamma$ for the four-bar, $\sigma = +1$ and -1 for the slider) and the sweep stays on the chosen branch; for a change-point linkage (parallelogram, deltoid) the circuit is named at a reference angle that does not depend on the sweep (the middle of the first interval between change points in 0 ... 2π, π/2 when there are none), so the label at a crank angle means the same motion on every turn; only the true change points (all four links collinear, $\theta_2 = 0$ or π) flip the branch, and it is followed by continuity through them inside the sweep. A linkage with one change point per turn (a kite) repeats every 720° and a sweep of more than one turn warns. Angles at which the linkage cannot be assembled, or at a toggle where the velocity system is singular, are left out of the table and counted. (Norton, Design of Machinery, position analysis of the four-bar and the offset slider-crank.)
A3. Input torque is by virtual work from the output load and the inertia of the output element only (four-bar: the output link's moment of inertia about its pivot; slider: the slider mass); the crank and the coupler or connecting rod are massless. (No link-inertia or gravity terms: add the link masses to the output element for a first allowance.)
A4. The peaks (speeds, accelerations, torque, transmission angle, swing) are those of the continuous motion over the swept range, found by a dense scan refined by golden section; the printed table (steps N, up to 3600) only samples it. (Peaks do not depend on the table's step count.)
A5. Cam: a translating follower on a cam turning at constant speed; the follower mass, a linear return spring (rate and preload at the base circle, for example linked from M7) and a constant external load make the force the cam must apply, $F = m \cdot a + k \cdot s + F_{pre} + F_{ext}$; contact is lost where it falls below zero. The segment table must add to 360° and close; pressure angle and undercut are checked for the roller follower ($R_p = R_b + R_r$), the positive profile radius for the flat-faced one. (Norton, Design of Machinery, cam design: the four classical laws, pressure angle and radius of curvature.)
A6. Follower force (four-bar: the tangential force at the output pin, $(T_L + J \cdot \alpha_4)/c$; slider: the slide-direction force $m \cdot a + F_L$; cam: the axial force) is the force the mechanism must apply to the output element. The force in the coupler ($F_{pin,pk}$) is the tangential force over sin μ; in the connecting rod it is the slide-direction force over cos θ_3, and the guide reaction is that force times tan θ_3; the roller contact normal force is F/cos φ. Crank, coupler and rod are massless two-force members. (Rigid body, frictionless.)
A7. When the sweep meets an assembly limit (a toggle of a four-bar, the rod perpendicular to the slide line of a slider-crank) the sweep is cut there and, because the speed, acceleration, torque and forces at constant input speed are unbounded at the limit, they are not reported; the limit angles are listed. A change-point four-bar (parallelogram, deltoid) is continued through its singular position on the same branch by continuity (γ changes sign there), but its speeds and torques next to the singular position are not reported. (Closed-form limit angles: $\ell = b + c$ or $|b - c|$ (four-bar), $|r \cdot \sin\theta - e| = l$ (slider).)
A8. Into F1 and E1: output_motion is a kinematic proxy at constant crank speed, not the cylinder speed of an actuated linkage. For E1 use the peak input_torque for the peak check and T_rms_in with n_in for the thermal sizing; the mean power is P_mean. (Chains 4 and 5, Addendum B §B7.)
A9. steps (Intervals N of the table (N + 1 rows, a whole number, at most 3600; the peaks do not depend on it)) is taken as 72. (5° table step.)
A10. mu_req (Minimum transmission angle required (linkages)) is taken as 40 deg. (Default minimum

transmission angle; set it from your design rule.)
A11. load_dir (External load: constant direction, or always opposing the output motion) is taken as Opposes motion. (The flap resists the blade in whichever direction it moves.)
A12. T_load (External load torque on the output link (four-bar). Positive acts in the $-\theta_4$ direction, resisting positive ω_4; with 'Opposes motion' its sign is applied against the actual direction of ω_4 (a negative value then assists)) is taken as 4 N*m. (Example flap-folding torque at the rocker shaft; measure it on the station.)
A13. J_out (Moment of inertia of the output link about its pivot (four-bar)) is taken as 0.012 kg*m^2. (Example: rocker arm and blade about the rocker pivot; take it from the CAD mass properties.)

4. Working

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[10] f_in = n_in / 60
= rpm to Hz
= 1 Hz
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Input (crank) frequency, for the spring surge check of M7 (mech.f_in)

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[16] S_plus_L = s + l                                          Grashof criterion
= shortest + longest link
= 195 mm
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Shortest plus longest link

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[17] P_plus_Q = p + q                                          Grashof criterion
= the other two links
= 240 mm
```
Sum of the other two links

```
[18] theta2_sw = θ₂                                Vector-loop closed form (Freudenstein / law of cosines)
= 73 rows, θ₂ from 0° to 360°
= [0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, … (73 items)] deg
```
Table: crank angle

```
[19] theta3_sw = θ₃ = atan2(B - A)                 Vector-loop closed form (Freudenstein / law of cosines)
= 73 rows, θ₂ from 0° to 360°
= [45.45, 43.305, 41.183, 39.119, 37.143, 35.277, 33.533, 31.922, 30.446, 29.103, 27.892, 26.
items)] deg
```
Table: coupler angle

```
[20] theta4_sw = θ₄ = □ □ □                        Vector-loop closed form (Freudenstein / law of cosines)
= 73 rows, θ₂ from 0° to 360°
= [93.891, 91.886, 90.182, 88.806, 87.774, 87.09, 86.749, 86.738, 87.038, 87.625, 88.475, 89.
items)] deg
```
Table: rocker angle

```
[21] omega3_sw = ω₃                                Vector-loop closed form (Freudenstein / law of cosines)
= 73 rows, θ₂ from 0° to 360°
= [-2.6928, -2.6892, -2.6368, -2.5435, -2.4183, -2.2706, -2.1088, -1.9402, -1.7705, -1.6037,
-1.2886, … (73 items)] rad/s
```
Table: coupler angular velocity

```
[22] omega4_sw = ω₄                                Vector-loop closed form (Freudenstein / law of cosines)
= 73 rows, θ₂ from 0° to 360°
= [-2.6928, -2.3371, -1.9398, -1.5152, -1.0781, -0.64174, -0.21757, 0.18575, 0.56219, 0.90807
1.5026, … (73 items)] rad/s
```
Table: rocker angular velocity

```
[23] alpha3_sw = α₃                                Vector-loop closed form (Freudenstein / law of cosines)
= 73 rows, θ₂ from 0° to 360°
= [-1.6439, 2.0963, 5.3499, 7.9793, 9.9336, 11.238, 11.97, 12.238, 12.157, 11.832, 11.354, 10
items)] rad/s²
```
Table: coupler angular acceleration

[24] alpha4_sw = α_4 Vector-loop closed form (Freudenstein / law of cosines)
= 73 rows, θ_2 from 0° to 360°
= [23.794, 27.273, 29.765, 31.194, 31.596, 31.098, 29.878, 28.129, 26.035, 23.751, 21.398, 19
items)] rad/s²
Table: rocker angular acceleration
[25] mu_sw = μ Vector-loop closed form (Freudenstein / law of cosines)
= 73 rows, θ_2 from 0° to 360°
= [48.441, 48.581, 48.999, 49.687, 50.631, 51.814, 53.216, 54.816, 56.592, 58.522, 60.583, 62
items)] deg
Table: transmission angle
[26] T_in_sw = (T_L + J·α_4)·α_4/α_2 Virtual work
= 73 rows, θ_2 from 0° to 360°
= [1.5919, 1.3661, 1.1246, 0.87436, 0.62126, 0.37043, 0.1261, 0.12823, 0.38586, 0.61929, 0.82
… (73 items)] N·m
Table: input torque by virtual work
[27] theta3_0 = θ_3 Vector-loop closed form (Freudenstein / law of cosines)
= at the first input angle
= 45.45 deg
Coupler angle at the first input angle
[28] theta4_0 = θ_4 Vector-loop closed form (Freudenstein / law of cosines)
= at the first input angle
= 93.891 deg
Rocker angle at the first input angle
[29] omega3_0 = ω_3 Vector-loop closed form (Freudenstein / law of cosines)
= at the first input angle
= -2.6928 rad/s
Coupler angular velocity at the first input angle
[30] omega4_0 = ω_4 Vector-loop closed form (Freudenstein / law of cosines)
= at the first input angle
= -2.6928 rad/s
Rocker angular velocity at the first input angle
[31] alpha3_0 = α_3 Vector-loop closed form (Freudenstein / law of cosines)
= at the first input angle
= -1.6439 rad/s²
Coupler angular acceleration at the first input angle
[32] alpha4_0 = α_4 Vector-loop closed form (Freudenstein / law of cosines)
= at the first input angle
= 23.794 rad/s²
Rocker angular acceleration at the first input angle
[33] mu_0 = μ Vector-loop closed form (Freudenstein / law of cosines)
= at the first input angle
= 48.441 deg
Transmission angle at the first input angle
[34] T_in_0 = (T_L + J·α_4)·α_4/α_2 Virtual work
= at the first input angle
= 1.5919 N·m
Input torque at the first input angle
[35] theta4_alt = θ_4 on the other circuit Vector-loop closed form (Freudenstein / law of cosines)
= mirror branch, at the first input angle
= 266.11 deg
Rocker angle on the other circuit at the first input angle
[36] theta4_min = min θ_4 Vector-loop closed form (Freudenstein / law of cosines)
= unwrapped along the sweep
= 86.704 deg

Smallest rocker angle

[37] theta4_max = max θ₄ — Vector-loop closed form (Freudenstein / law of cosines)
= unwrapped along the sweep
= 141.49 deg

Largest rocker angle

[38] swing = max θ₄ - min θ₄ — Vector-loop closed form (Freudenstein / law of cosines)
= over the sweep
= 54.786 deg

Rocker swing over the sweep

[39] mu_min = min μ — Vector-loop closed form (Freudenstein / law of cosines)
= dense scan, refined
= 48.441 deg

Smallest transmission angle over the sweep

[40] res_max = max |loop residual| — Vector-loop closed form (Freudenstein / law of cosines)
= 73 table rows and 1441 scan angles
= 9.3095e-14 mm

Largest loop-closure residual

[41] res_vel_max = max |velocity loop| / its terms — Vector-loop closed form (Freudenstein / law of cosines)
= 73 table rows
= 2.1412e-16

Largest velocity-loop residual, relative to the size of its terms

[42] omega3_pk = max |ω₃| — Vector-loop closed form (Freudenstein / law of cosines)
= dense scan, refined
= 2.6976 rad/s

Peak coupler angular velocity

[43] omega_out_pk = max |ω₄| — Vector-loop closed form (Freudenstein / law of cosines)
= dense scan, refined
= 3.5435 rad/s

Peak output (rocker) angular velocity

[44] alpha3_pk = max |α₃| — Vector-loop closed form (Freudenstein / law of cosines)
= dense scan, refined
= 22.724 rad/s²

Peak coupler angular acceleration

[45] alpha_out_pk = max |α₄| — Vector-loop closed form (Freudenstein / law of cosines)
= dense scan, refined
= 31.599 rad/s²

Peak output (rocker) angular acceleration

[46] input_torque = max |T_in| — Virtual work
= dense scan, refined
= 2.2654 N·m

Peak input (crank) torque by virtual work (mech.input_torque → E1 peak torque)

[47] output_motion = c · max |ω₄| — Vector-loop closed form (Freudenstein / law of cosines)
= output pin at the rocker length
= 0.35435 m/s

Peak speed of the output pin at the rocker length c (mech.output_motion; a kinematic proxy for F1 only if the cylinder acts perpendicular to the rocker at that radius; the angular velocity is omega_out_pk)

[48] peak_accel = max |a_pin| — Vector-loop closed form (Freudenstein / law of cosines)
= √(α₄² + ω₄⁴)·c
= 3.1625 m/s²

Peak acceleration of the output pin (mech.peak_accel)

[49] follower_force = max |(T_L + J·α₄)/c| — Virtual work
= tangential force at the output pin
= 43.476 N

Peak force perpendicular to the rocker at the output pin, the drive force of the output (mech.follower_force)

[50] F_pin_pk = max |(T_L + J·□₄)/(c·sin □)| Virtual work
= force in the coupler = the crank-pin force
= 53.69 N
Peak force in the coupler, which is also the force at the crank pin and the pin loads on the bearings (mech.F_pin_pk → M4, M5)
[51] T_rms_in = √(mean T_in²) Virtual work
= Simpson over the scan
= 1.3744 N·m
RMS input torque over the swept range (mech.T_rms_in → E1 thermal sizing)
[52] P_mean = □·mean(T_in) Virtual work
= Simpson over the scan
= 7.6496 W
Mean input power over the swept range (negative: the mechanism returns energy) (mech.P_mean)
[53] u_mu = □_req / □_min Transmission angle
= 40° / 48.441°
= 0.82575
Transmission angle: required minimum against the smallest over the sweep

5. Results

SYMBOL	VALUE	DESCRIPTION
n_in	60 rpm	Input (crank) speed, passed through for E1's load speed (mech.n_in)
f_in	1 Hz	Input (crank) frequency, for the spring surge check of M7 (mech.f_in)
grashof	"Grashof crank-rocker, input link shortest"	Grashof classification
S_plus_L	195 mm	Shortest plus longest link
P_plus_Q	240 mm	Sum of the other two links
theta2_sw	[0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, … (73 items)] deg	Table: crank angle
theta3_sw	[45.45, 43.305, 41.183, 39.119, 37.143, 35.277, 33.533, 31.922, 30.446, 29.103, 27.892, 26.806, … (73 items)] deg	Table: coupler angle
theta4_sw	[93.891, 91.886, 90.182, 88.806, 87.774, 87.09, 86.749, 86.738, 87.038, 87.625, 88.475, 89.561, … (73 items)] deg	Table: rocker angle
omega3_sw	[-2.6928, -2.6892, -2.6368, -2.5435, -2.4183, -2.2706, -2.1088, -1.9402, -1.7705, -1.6037, -1.4425, -1.2886, … (73 items)] rad/s	Table: coupler angular velocity

SYMBOL	VALUE	DESCRIPTION
omega4_sw	[-2.6928, -2.3371, -1.9398, -1.5152, -1.0781, -0.64174, -0.21757, 0.18575, 0.56219, 0.90807, 1.2216, 1.5026, … (73 items)] rad/s	Table: rocker angular velocity
alpha3_sw	[-1.6439, 2.0963, 5.3499, 7.9793, 9.9336, 11.238, 11.97, 12.238, 12.157, 11.832, 11.354, 10.795, … (73 items)] rad/s²	Table: coupler angular acceleration
alpha4_sw	[23.794, 27.273, 29.765, 31.194, 31.596, 31.098, 29.878, 28.129, 26.035, 23.751, 21.398, 19.064, … (73 items)] rad/s²	Table: rocker angular acceleration
mu_sw	[48.441, 48.581, 48.999, 49.687, 50.631, 51.814, 53.216, 54.816, 56.592, 58.522, 60.583, 62.755, … (73 items)] deg	Table: transmission angle
T_in_sw	[1.5919, 1.3661, 1.1246, 0.87436, 0.62126, 0.37043, 0.1261, 0.12823, 0.38586, 0.61929, 0.82764, 1.0113, … (73 items)] N·m	Table: input torque by virtual work
theta3_0	45.45 deg	Coupler angle at the first input angle
theta4_0	93.891 deg	Rocker angle at the first input angle
omega3_0	-2.6928 rad/s	Coupler angular velocity at the first input angle
omega4_0	-2.6928 rad/s	Rocker angular velocity at the first input angle
alpha3_0	-1.6439 rad/s²	Coupler angular acceleration at the first input angle
alpha4_0	23.794 rad/s²	Rocker angular acceleration at the first input angle
mu_0	48.441 deg	Transmission angle at the first input angle
T_in_0	1.5919 N·m	Input torque at the first input angle
theta4_alt	266.11 deg	Rocker angle on the other circuit at the first input angle
theta4_min	86.704 deg	Smallest rocker angle
theta4_max	141.49 deg	Largest rocker angle
swing	54.786 deg	Rocker swing over the sweep
mu_min	48.441 deg	Smallest transmission angle over the sweep
res_max	9.3095e-14 mm	Largest loop-closure residual

SYMBOL	VALUE	DESCRIPTION
res_vel_max	2.1412e-16	Largest velocity-loop residual, relative to the size of its terms
toggle_angles	"none: coupler and rocker are never collinear"	Toggle crank angles (coupler and rocker collinear, μ = 0): the limits of the crank's assemblable range
limit_positions	"θ₂ = 32.66° with θ₄ = 86.704°; θ₂ = 220.95° with θ₄ = 141.49°"	Crank and coupler collinear (the rocker's dead-centre positions)
omega3_pk	2.6976 rad/s	Peak coupler angular velocity
omega_out_pk	3.5435 rad/s	Peak output (rocker) angular velocity
alpha3_pk	22.724 rad/s²	Peak coupler angular acceleration
alpha_out_pk	31.599 rad/s²	Peak output (rocker) angular acceleration
input_torque	2.2654 N·m	Peak input (crank) torque by virtual work (mech.input_torque → E1 peak torque)
output_motion	0.35435 m/s	Peak speed of the output pin at the rocker length c (mech.output_motion; a kinematic proxy for F1 only if the cylinder acts perpendicular to the rocker at that radius; the angular velocity is omega_out_pk)
peak_accel	3.1625 m/s²	Peak acceleration of the output pin (mech.peak_accel)
follower_force	43.476 N	Peak force perpendicular to the rocker at the output pin, the drive force of the output (mech.follower_force)
F_pin_pk	53.69 N	Peak force in the coupler, which is also the force at the crank pin and the pin loads on the bearings (mech.F_pin_pk → M4, M5)
T_rms_in	1.3744 N·m	RMS input torque over the swept range (mech.T_rms_in → E1 thermal sizing)
P_mean	7.6496 W	Mean input power over the swept range (negative: the mechanism returns energy) (mech.P_mean)
u_mu	0.82575	Required against smallest transmission angle

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Transmission angle μ_min at or above the required minimum	u_mu = 0.82575	u_mu ≤ 1	0.826	Transmission angle
PASS	Loop-closure residual below 1e-9 mm at every step	max residual = 9.3095e-14 mm	< 1e-9 mm	0	Self-consistency
PASS	Velocity-loop residual below 1e-8 of the size of its terms	max residual = 2.1412e-16	< 1e-8	0	Self-consistency

7. Applicability

No warnings.

8. Verification

Template mech.mechanism@1.2.0 Content hash 082569fdde676480 Engine calc-engine/1.0.0
Known-answer cases passed 16/16 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0